gaming_academic_performance_eda_project

May 31, 2026

0.1 Impact of Gaming Habits on Academic Performance

0.2 Problem Statement

0.3 The objective of this project is to analyze how gaming habits, lifestyle patterns, stress levels, and study behavior affect students' academic performance using exploratory data analysis techniques in Python.

0.4 Import Libraries

[1]:
```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
```

0.5 Load Dataset

[2]:
```
data = pd.read_csv(r"C:\Gaming Dataset\Gaming_Academic_Performance.csv")
```

[3]:
```
data.head(10)
```

[3]:

	student_id	age	gender	gaming_hours	study_hours	sleep_hours	\
0	1	22	Male	7.2	8.8	7.0	
1	2	19	Male	0.1	8.7	7.6	
2	3	23	Female	1.7	9.6	4.4	
3	4	20	Female	6.6	1.7	7.8	
4	5	22	Female	5.4	5.8	5.6	
5	6	18	Male	7.2	6.8	8.0	
6	7	22	Female	0.5	7.4	6.6	
7	8	23	Female	4.8	6.8	5.4	
8	9	20	Female	4.6	7.9	8.8	
9	10	19	Male	0.3	4.2	7.8	

	attendance	gaming_genre	social_activity	device_usage	reaction_time_ms	\
0	91.4	FPS	3.3	9.4	235.84	
1	63.6	Casual	1.0	3.2	328.71	
2	83.3	Casual	3.5	5.6	313.61	
3	75.0	RPG	1.5	11.8	241.84	
4	65.6	FPS	1.0	8.2	249.31	

```
5         68.1          RPG                3.7           10.9              242.75
6         91.3       Casual                1.8            2.9              320.20
7         66.5          FPS                2.9           10.0              271.21
8         72.5          FPS                1.0            9.5              272.39
9         71.4       Casual                3.1            5.4              318.59

   addiction_score stress_level  grades
0            14.69          Low      86
1             2.47       Medium      98
2             4.73         High      91
3            14.54          Low      33
4            12.48          Low      59
5            18.75          Low      58
6             1.74       Medium      94
7            11.75       Medium      62
8            11.04       Medium      81
9             4.72       Medium      82
```

0.6 Basic Exploration

[4]:
```python
data.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 8000 entries, 0 to 7999
Data columns (total 14 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   student_id        8000 non-null   int64
 1   age               8000 non-null   int64
 2   gender            8000 non-null   str
 3   gaming_hours      8000 non-null   float64
 4   study_hours       8000 non-null   float64
 5   sleep_hours       8000 non-null   float64
 6   attendance        8000 non-null   float64
 7   gaming_genre      8000 non-null   str
 8   social_activity   8000 non-null   float64
 9   device_usage      8000 non-null   float64
 10  reaction_time_ms  8000 non-null   float64
 11  addiction_score   8000 non-null   float64
 12  stress_level      8000 non-null   str
 13  grades            8000 non-null   int64
dtypes: float64(8), int64(3), str(3)
memory usage: 875.1 KB
```

[5]:
```python
data.describe()
```

[5]:

	student_id	age	gaming_hours	study_hours	sleep_hours	\
count	8000.00000	8000.000000	8000.000000	8000.000000	8000.000000	
mean	4000.50000	19.983625	4.091337	5.465750	6.498100	
std	2309.54541	2.587072	2.309025	2.575802	1.443127	
min	1.00000	16.000000	0.000000	1.000000	4.000000	
25%	2000.75000	18.000000	2.100000	3.200000	5.200000	
50%	4000.50000	20.000000	4.100000	5.500000	6.500000	
75%	6000.25000	22.000000	6.100000	7.700000	7.700000	
max	8000.00000	24.000000	8.000000	10.000000	9.000000	

	attendance	social_activity	device_usage	reaction_time_ms	\
count	8000.000000	8000.000000	8000.000000	8000.000000	
mean	79.891225	2.512587	7.590975	271.105839	
std	11.580569	1.441520	2.709980	29.440675	
min	60.000000	0.000000	1.100000	183.260000	
25%	69.800000	1.300000	5.600000	247.160000	
50%	79.700000	2.500000	7.600000	270.475000	
75%	90.100000	3.800000	9.600000	294.690000	
max	100.000000	5.000000	14.000000	347.870000	

	addiction_score	grades
count	8000.000000	8000.000000
mean	9.922191	66.083875
std	5.006251	22.253904
min	0.000000	0.000000
25%	5.920000	50.000000
50%	10.005000	67.000000
75%	13.860000	84.000000
max	23.160000	100.000000

[6]:

```
data.shape
```

[6]: (8000, 14)

[7]:

```
data.isnull().sum()
```

[7]:

```
student_id          0
age                 0
gender              0
gaming_hours        0
study_hours         0
sleep_hours         0
attendance          0
gaming_genre        0
social_activity     0
device_usage        0
reaction_time_ms    0
```

```
addiction_score     0
stress_level        0
grades              0
dtype: int64
```

[114]:
```python
data.duplicated().sum()
```

[114]:
```
np.int64(0)
```

0.7 Feature Engineering

[8]:
```python
data["grades"] = np.where(data["grades"] > 100, 100, data["grades"]).round().
 ↪astype(int)
```

[9]:
```python
data["addiction_score"] = np.where(data["addiction_score"] < 0, 0,␣
 ↪data["addiction_score"]).round().astype(int)
```

0.8 Create Performance Category

[106]:
```python
# Creating performance categories based on grades

data["performance_level"] = pd.cut(
    data["grades"],
    bins=[-1,50,75,100],
    labels=["Low","Medium","High"]
)
```

0.9 Gaming Addiction Risk

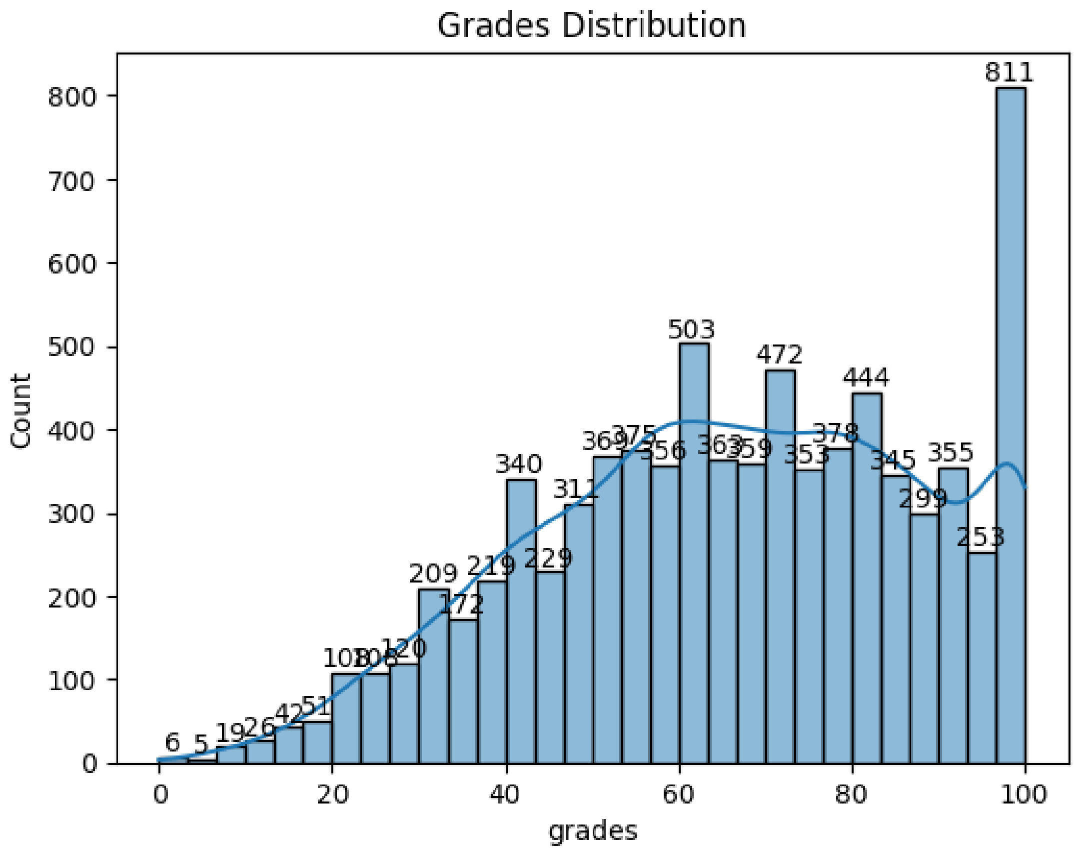
[107]:
```python
# Creating addiction_risk categories based on addiction_score

data["addiction_risk"] = pd.cut(
    data["addiction_score"],
    bins=[-1,10,20,30],
    labels=["Low","Medium","High"]
)
```

0.10 Sleep Quality

[108]:
```python
# Creating sleep_quality categories based on sleep_hours

data["sleep_quality"] = np.where(
    data["sleep_hours"] < 7,
    "Poor Sleep",
    "Healthy Sleep"
)
```

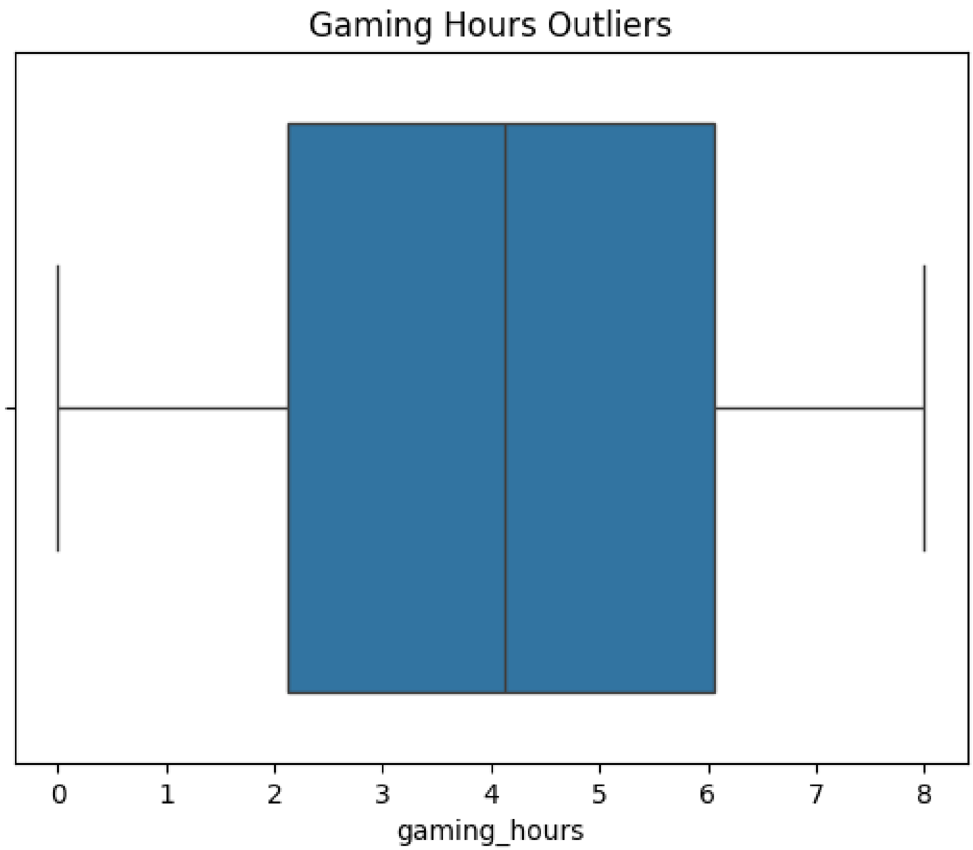
0.11 Grades Distribution

[115]:
```
ax = sns.histplot(data["grades"], kde=True)
ax.bar_label(ax.containers[0])
plt.title("Grades Distribution")
plt.show()
```



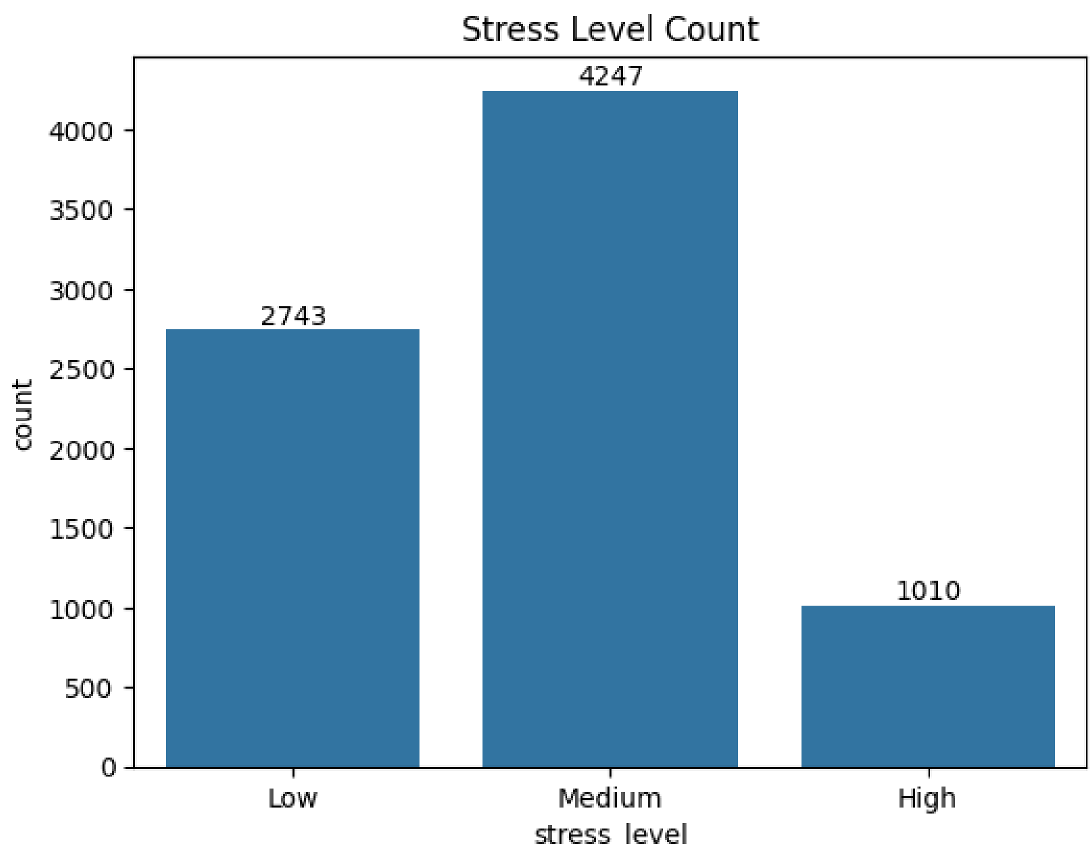
0.12 Gaming Hours

[116]:
```
sns.boxplot(x=data["gaming_hours"])
plt.title("Gaming Hours Outliers")
plt.show()
```



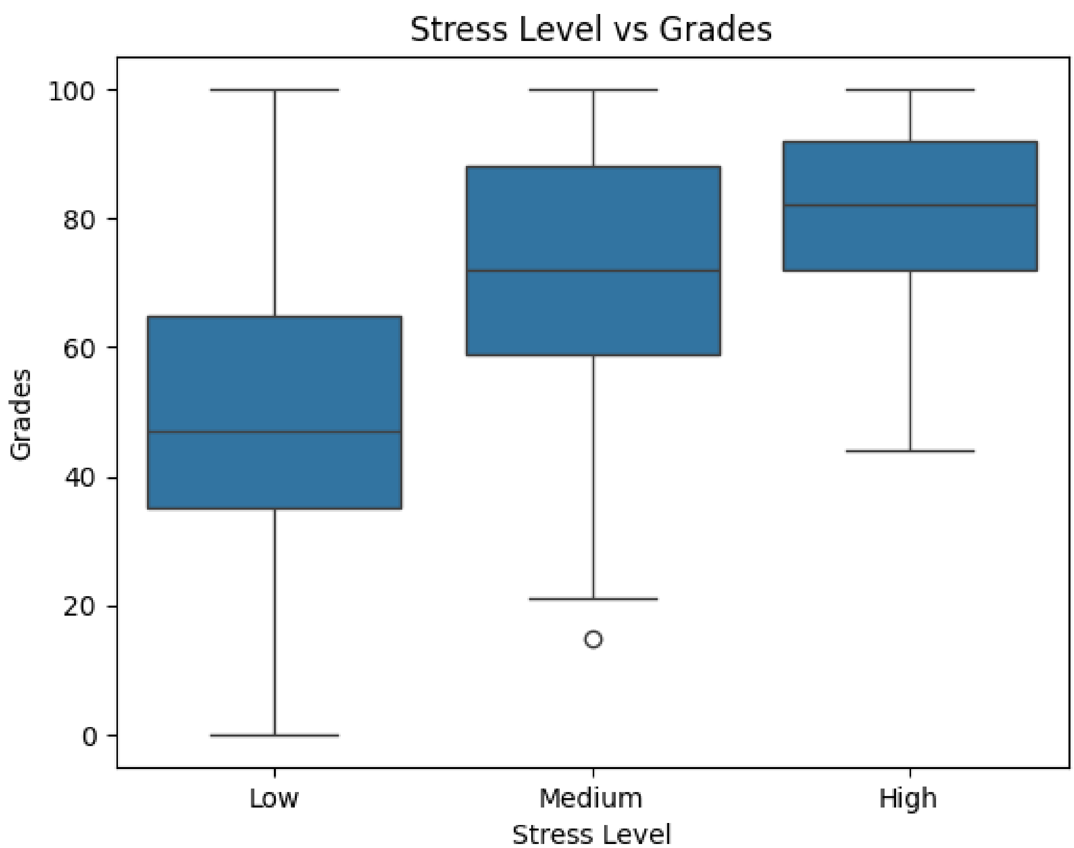
0.13 Stress Levels

[117]:
```
ax = sns.countplot(x=data["stress_level"])
ax.bar_label(ax.containers[0])
plt.title("Stress Level Count")
plt.show()
```



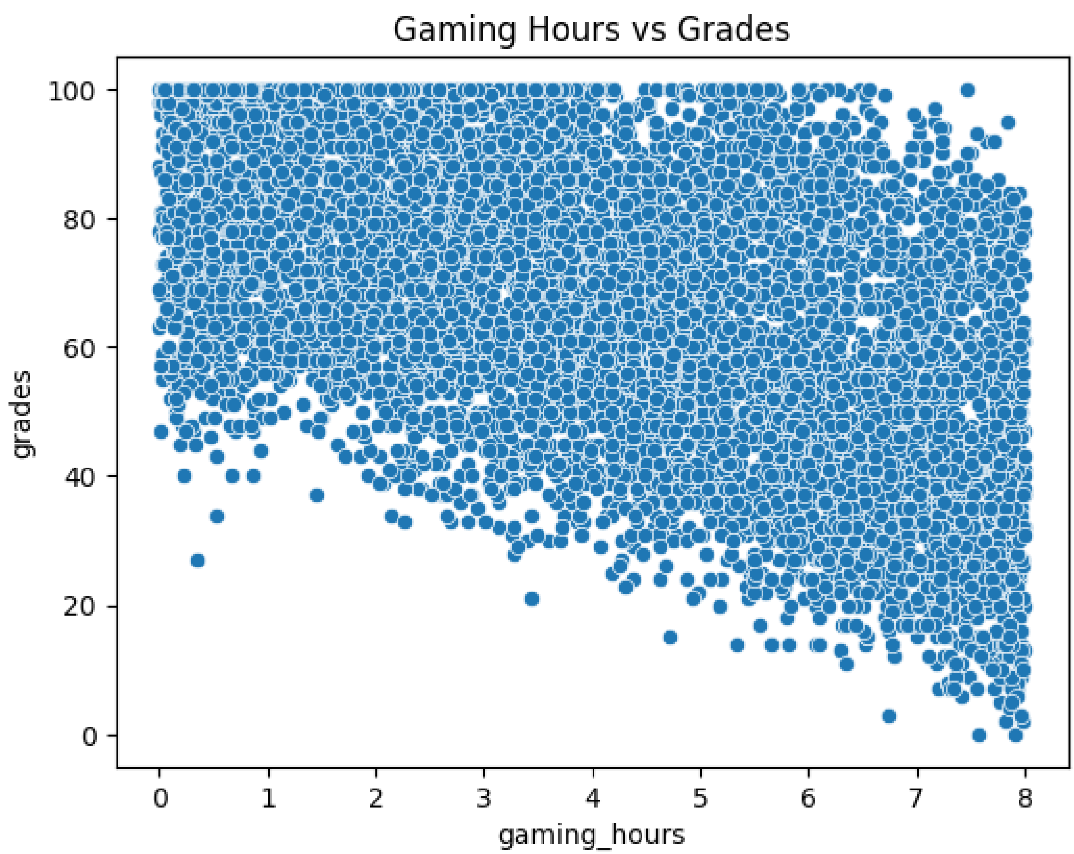
0.14 Stress Level vs Grades

[118]:
```
sns.boxplot(
    x="stress_level",
    y="grades",
    data=data
)

plt.title("Stress Level vs Grades")
plt.xlabel("Stress Level")
plt.ylabel("Grades")

plt.show()
```



0.15 Gaming Hours vs Grades

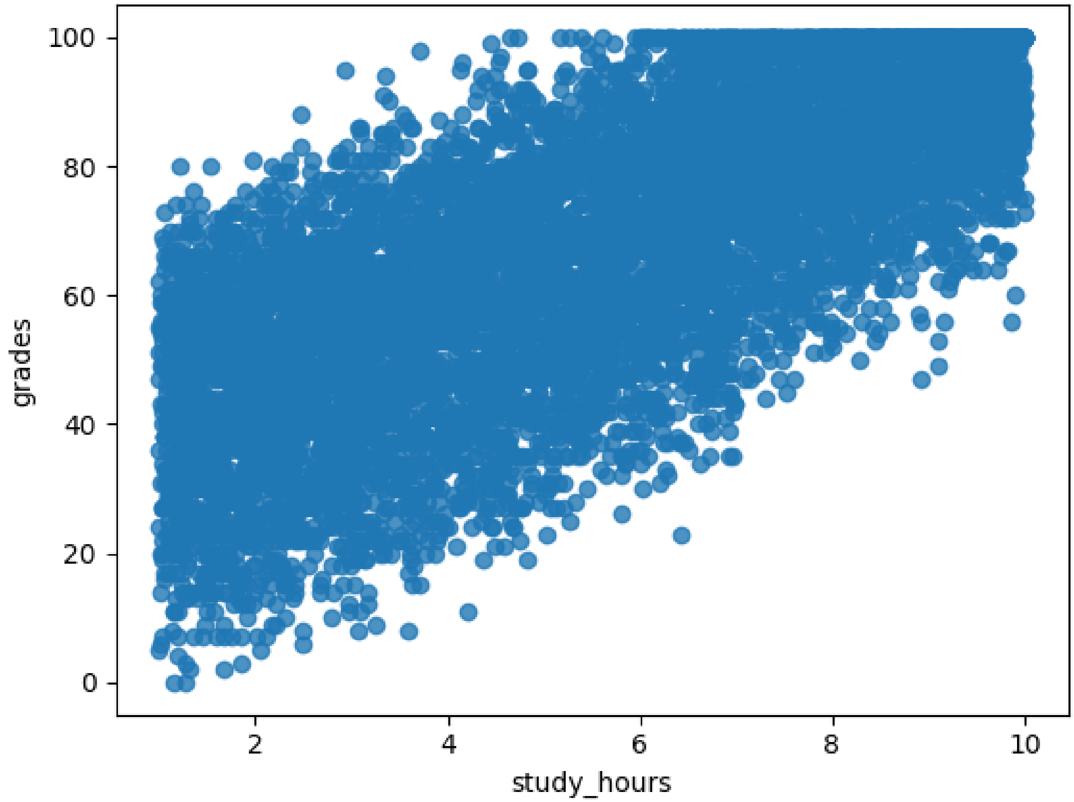
[119]:
```
sns.scatterplot(
    x="gaming_hours",
    y="grades",
    data=data
)

plt.title("Gaming Hours vs Grades")
plt.show()
```



0.16 Study Hours vs Grades

[121]:
```
sns.regplot(
    x="study_hours",
    y="grades",
    data=data
)
```

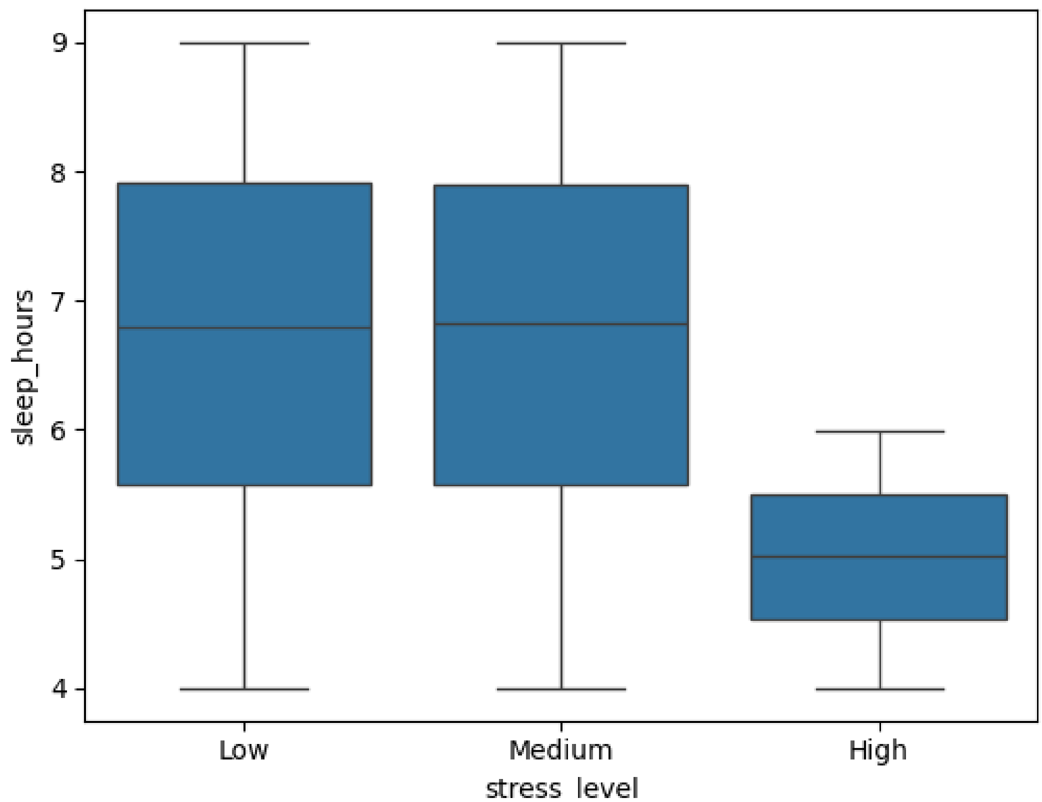
[121]: <Axes: xlabel='study_hours', ylabel='grades'>



0.17 Sleep vs Stress

[122]:
```
sns.boxplot(
    x="stress_level",
    y="sleep_hours",
    data=data
)
```

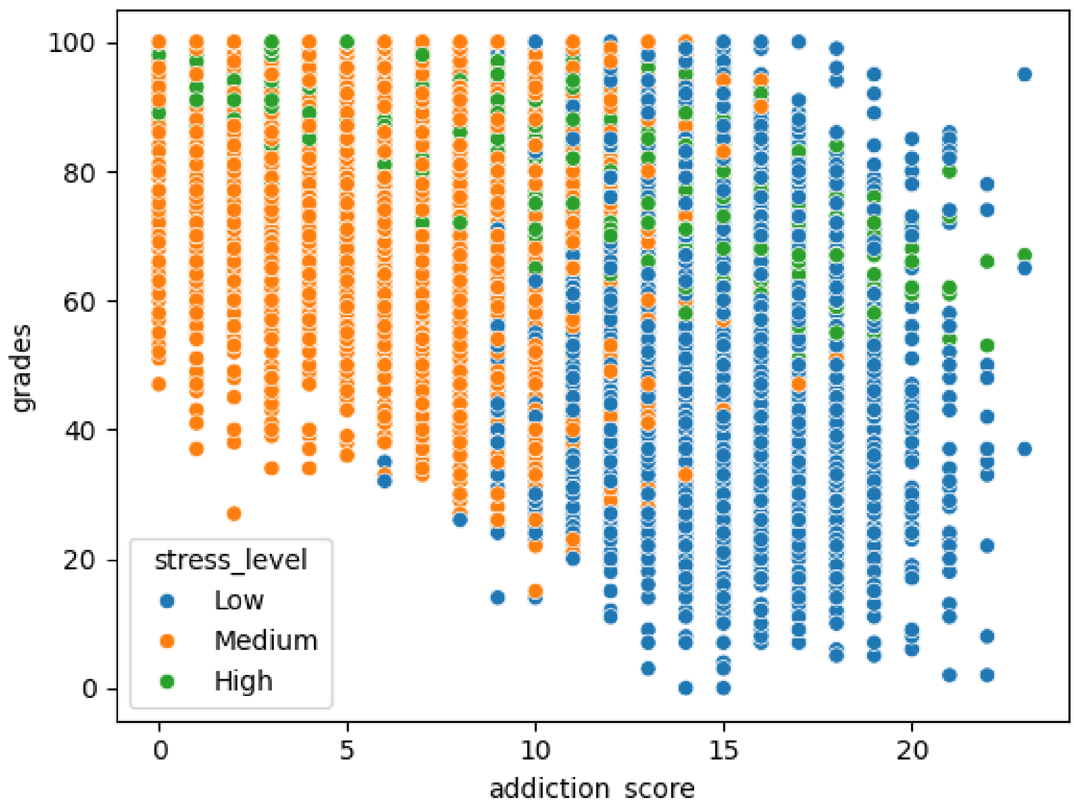
[122]: <Axes: xlabel='stress_level', ylabel='sleep_hours'>



0.18 Addiction Score vs Grades

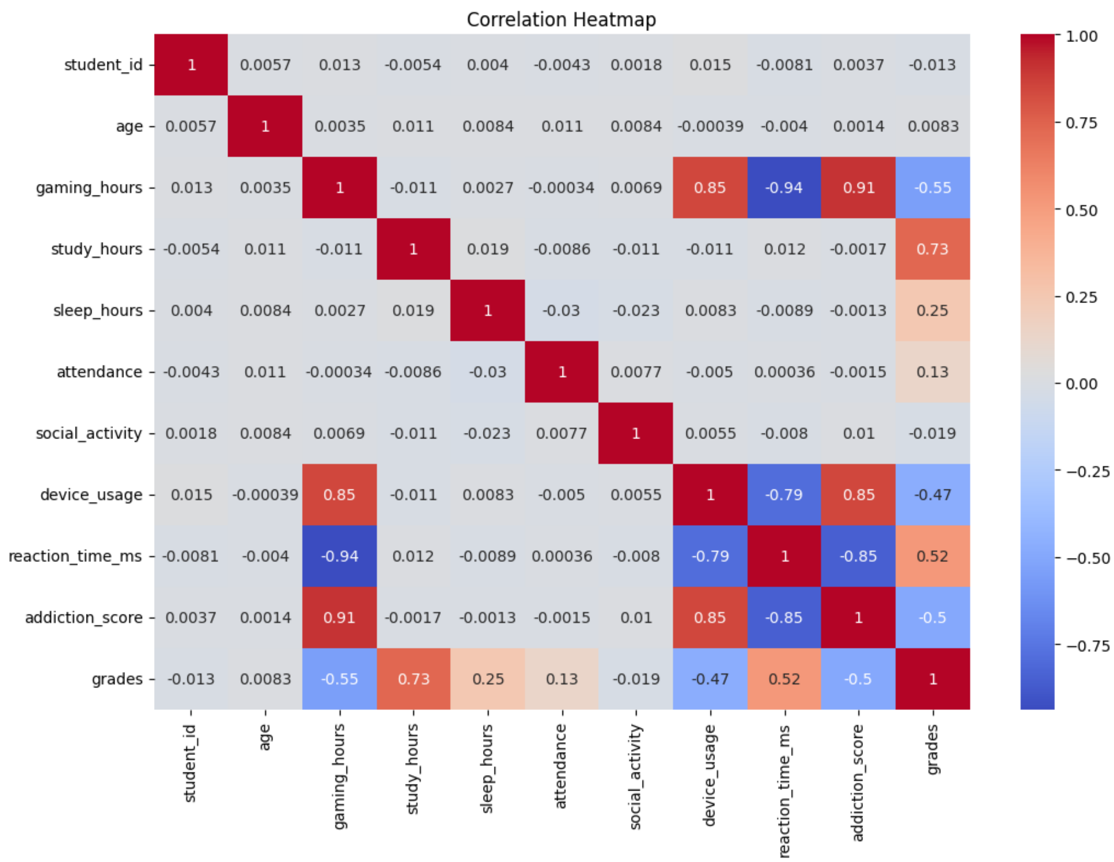
[124]:
```
sns.scatterplot(
    x="addiction_score",
    y="grades",
    hue="stress_level",
    data=data
)
```

[124]: <Axes: xlabel='addiction_score', ylabel='grades'>



0.19 Correlation Heatmap

[125]:
```
plt.figure(figsize=(12,8))

sns.heatmap(
    data.corr(numeric_only=True),
    annot=True,
    cmap="coolwarm"
)

plt.title("Correlation Heatmap")
plt.show()
```

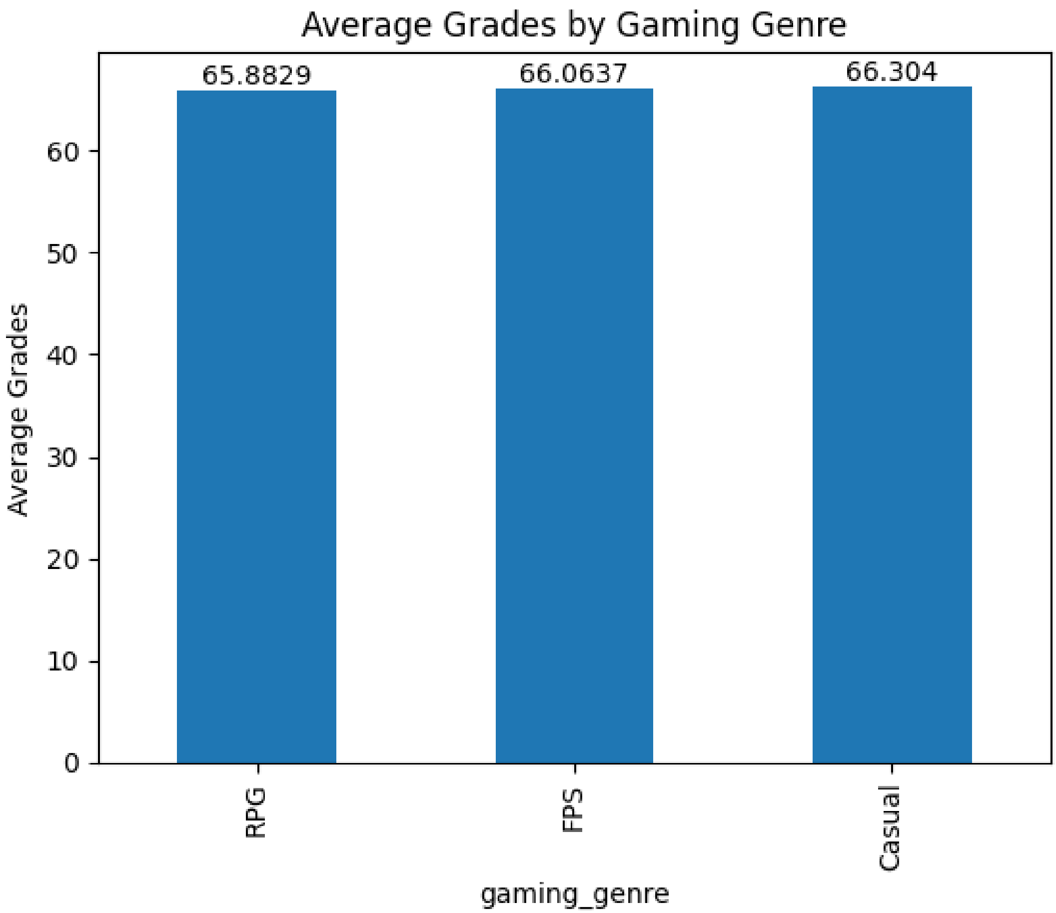


0.20 Correlation Summary

- Study hours and attendance positively correlate with grades.
- Gaming hours and addiction score negatively correlate with grades.
- Stress level is associated with lower academic performance.

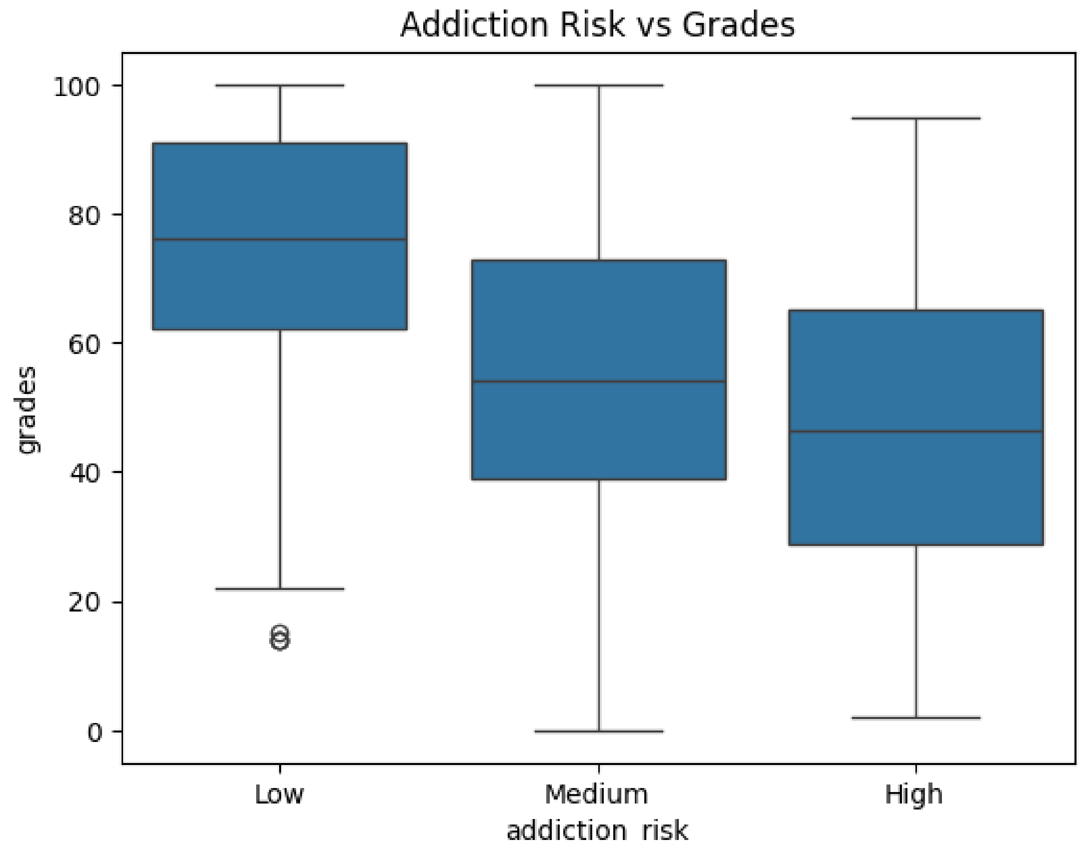
0.21 Average Grades by Gaming Genre

[126]:
```
genre_perf = data.groupby("gaming_genre")["grades"].mean().sort_values()

ax = genre_perf.plot(kind="bar")
ax.bar_label(ax.containers[0])
plt.title("Average Grades by Gaming Genre")
plt.ylabel("Average Grades")
plt.show()
```



0.22 Average Grades by Addiction Risk

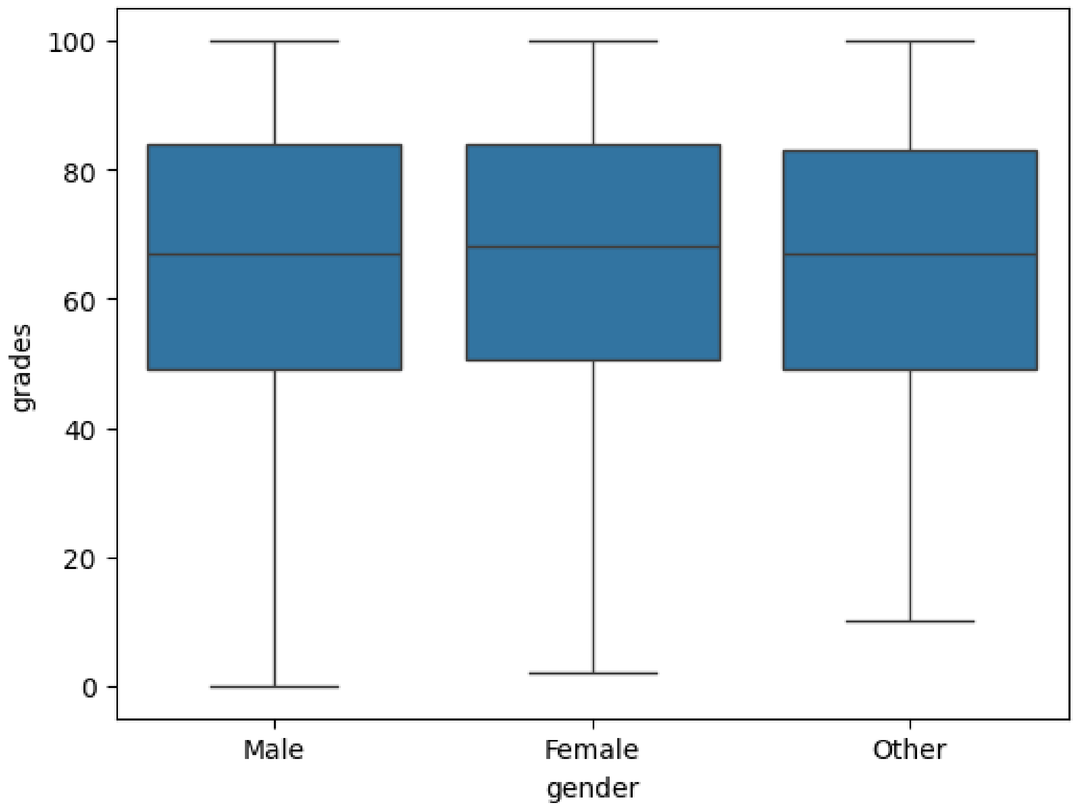
[127]:
```
sns.boxplot(
    x="addiction_risk",
    y="grades",
    data=data
)

plt.title("Addiction Risk vs Grades")
plt.show()
```



0.23 Gender-Based Analysis

[128]:
```
sns.boxplot(
    x="gender",
    y="grades",
    data=data
)
```

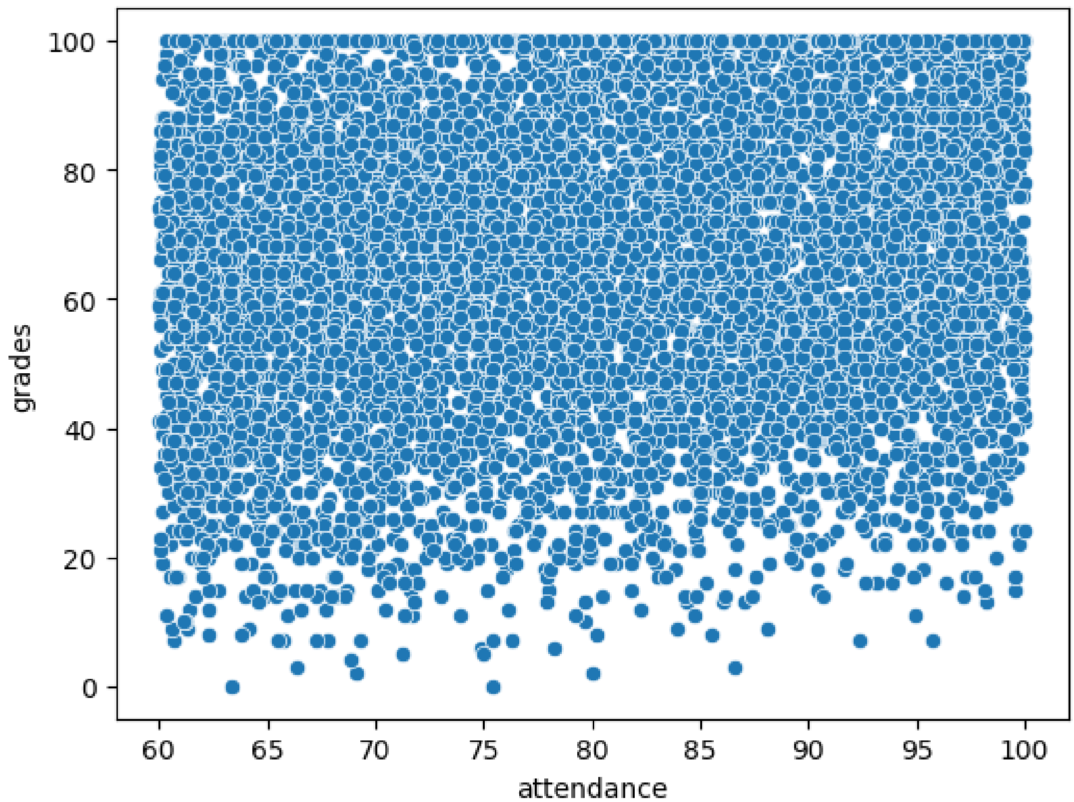
[128]: <Axes: xlabel='gender', ylabel='grades'>



0.24 Attendance vs Performance

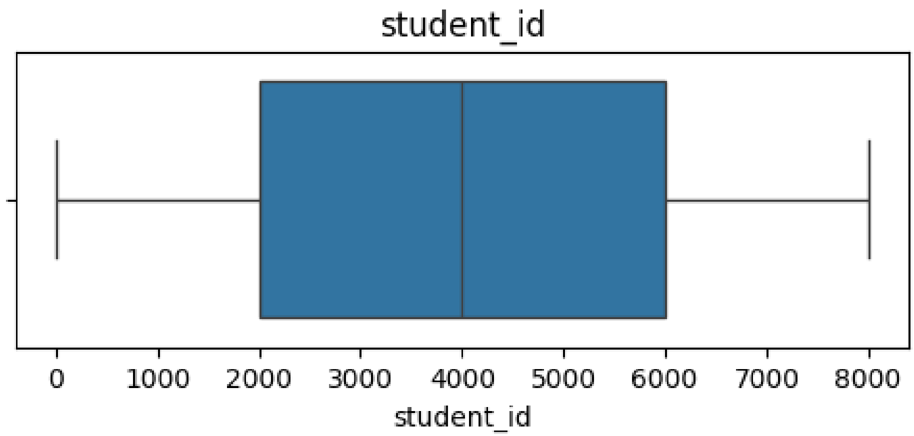
[129]:
```
sns.scatterplot(
    x="attendance",
    y="grades",
    data=data
)
```

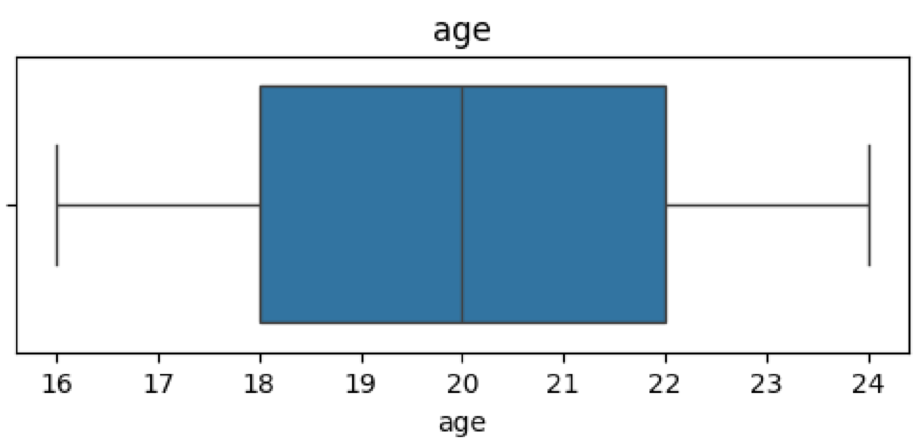
[129]: `<Axes: xlabel='attendance', ylabel='grades'>`



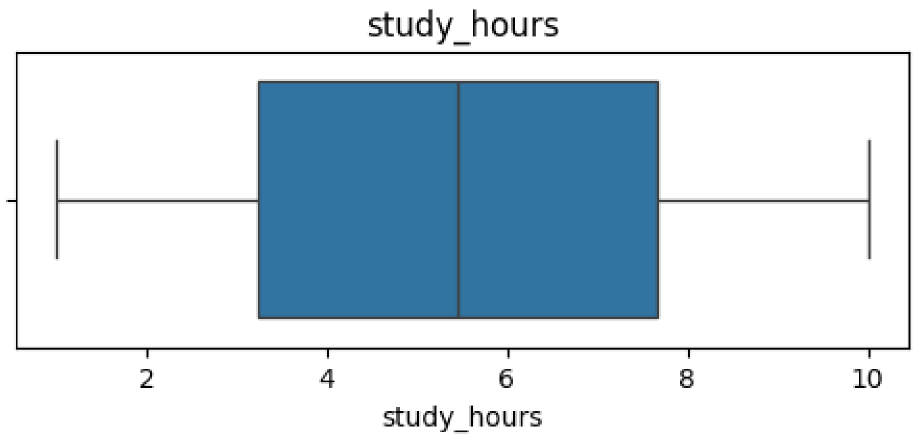
0.25 Outlier Detection

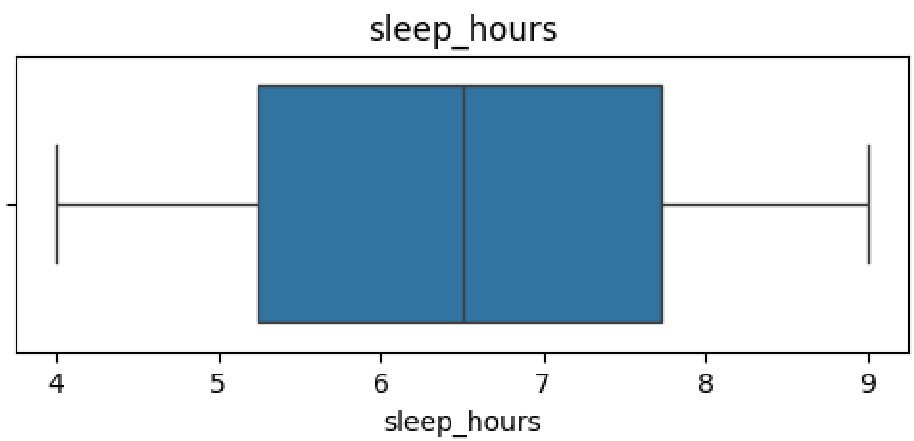
[131]:
```
numeric_cols = data.select_dtypes(include=np.number).columns

for col in numeric_cols:
    plt.figure(figsize=(6,2))
    sns.boxplot(x=data[col])
    plt.title(col)
    plt.show()
```

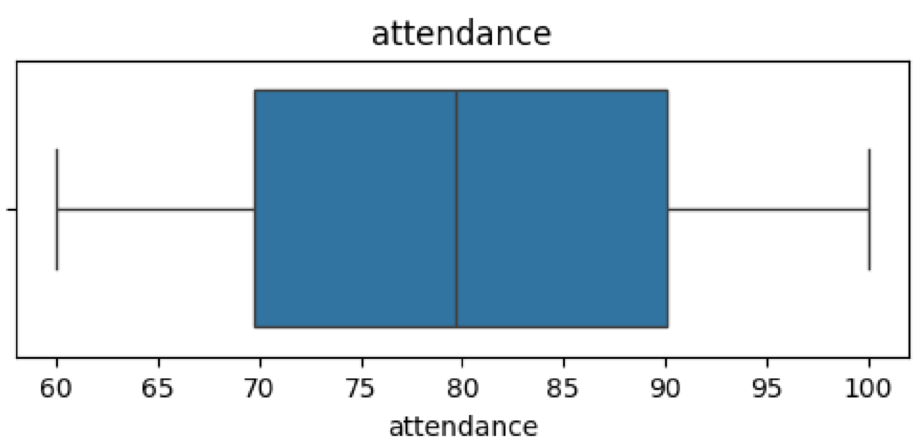

student_id
0
1000
2000
3000
4000
5000
6000
7000
8000
student_id

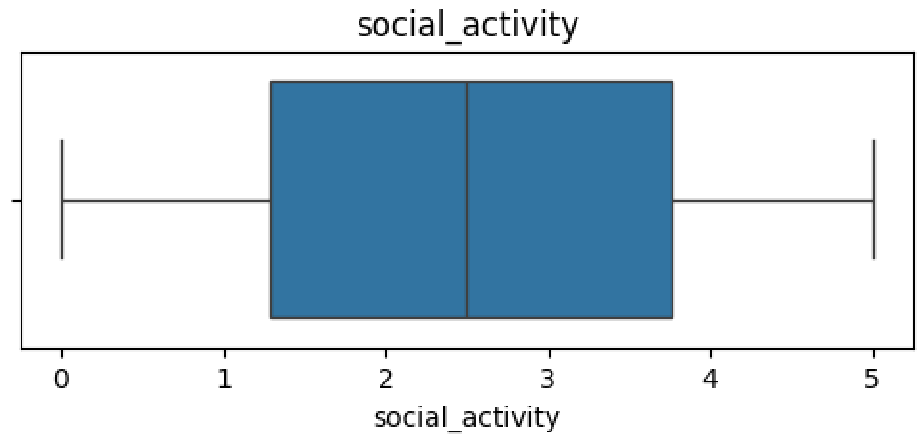

age
16
17
18
19
20
21
22
23
24
age

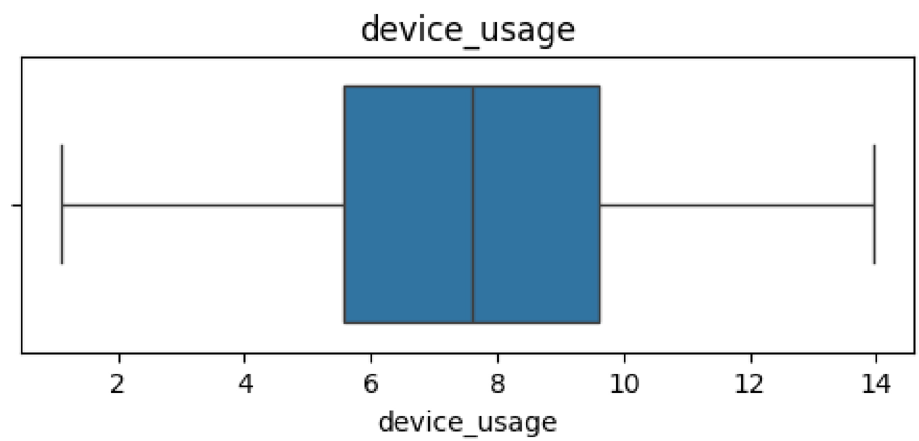

gaming_hours
0
1
2
3
4
5
6
7
8
gaming_hours

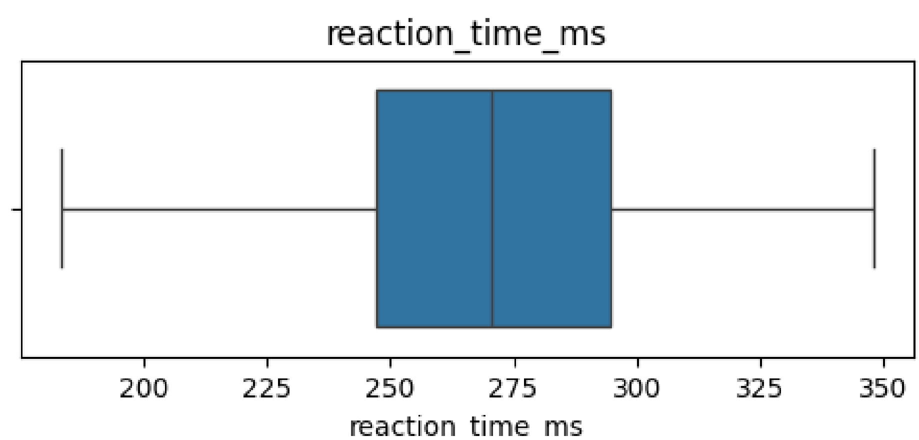

study_hours
2
4
6
8
10
study_hours

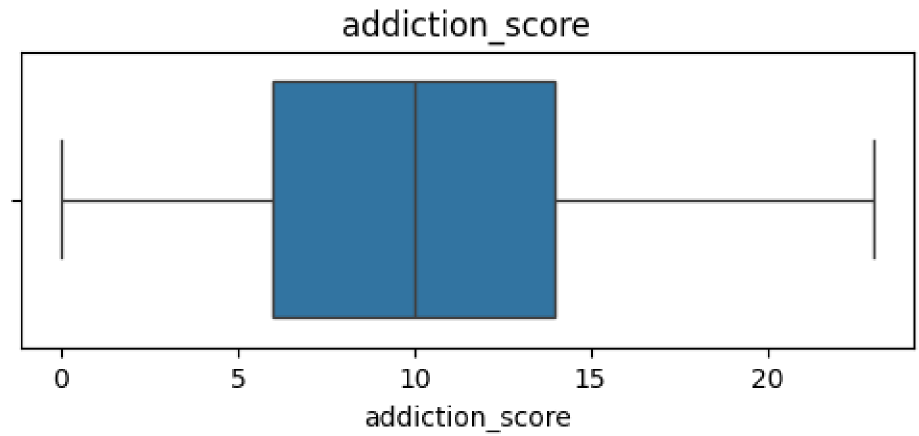

sleep_hours
4
5
6
7
8
9
sleep_hours

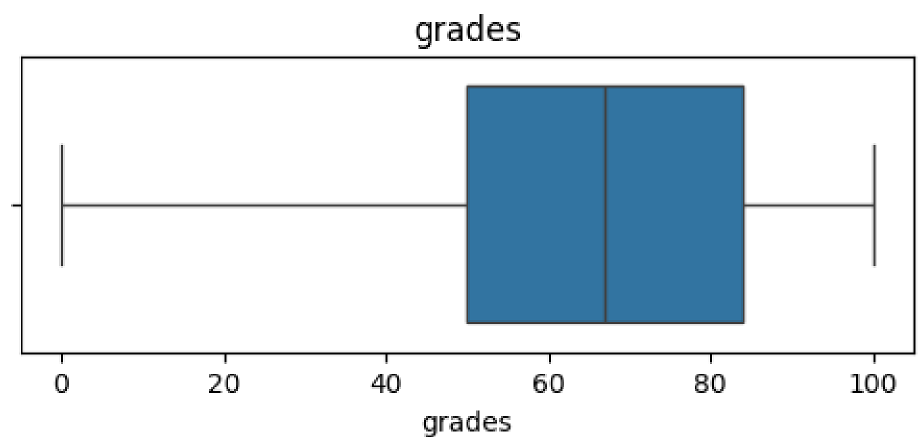

attendance
60
65
70
75
80
85
90
95
100
attendance


social_activity
0
1
2
3
4
5
social_activity


device_usage
2
4
6
8
10
12
14
device_usage


reaction_time_ms
200
225
250
275
300
325
350
reaction_time_ms





0.26 Key Insights

- Students with high gaming hours tend to score lower grades.
- Study hours positively impact academic performance.
- High addiction scores are associated with increased stress levels.
- Students with healthy sleep patterns generally perform better academically.
- Attendance shows strong positive correlation with grades.
- High stress levels negatively affect academic performance.
- Excessive device usage may contribute to reduced study efficiency.

0.27 Conclusion

This analysis explored the relationship between gaming behavior, lifestyle habits, and academic performance among students.

The findings suggest that balanced gaming habits, proper sleep, regular attendance, and effective study routines contribute positively to academic success. On the other hand, excessive gaming and high stress levels may negatively impact student performance.

This project demonstrates the use of Python for data cleaning, exploratory data analysis, visualization, and behavioral insight generation.

0.28 Recommendations

- Encourage balanced gaming habits among students.
- Promote healthy sleep schedules and stress management.
- Monitor students with high addiction risk scores.
- Increase awareness regarding excessive screen time.
- Provide academic support to low-attendance students.

1 Future Scope

- Build a machine learning model to predict student grades.
- Develop an interactive Power BI dashboard.
- Perform clustering analysis for student segmentation.
- Apply statistical testing for deeper behavioral analysis.